

GeoFare – 1. Professional Coding Standards and Practices

Software Engineering Activity: Identification of code that implements and does not fully implement Professional Coding Standards and Practices.

1. Meaning of Professional Coding Standards and Practices

Professional coding standards promote readable, consistent, maintainable, secure, and organized source code. The assessment below is based only on the GeoFare Kotlin files provided.

2. Code That Implements the Standards

File / Code	Finding	Assessment
HomeScreen.kt	Uses clear names such as userName , menuItems , MenuItem , and firestore . The UI is organized into composable functions.	Implements
MainActivity.kt	Separates application setup into GeoFareApp() and uses a clear navigation structure with named routes.	Implements
LoginScreen.kt	Uses descriptive state variables such as email , password , isLoading , and errorMessage .	Implements
RegisterScreen.kt	Uses meaningful variables and separates registration UI from authentication/database operations.	Implements
SplashScreen.kt	Uses a dedicated composable for the splash screen and descriptive animation variables such as scale .	Implements
FareCalculationScreen.kt	Uses a separate screen function and clear navigation handling.	Implements
TripHistoryScreen.kt	Uses a dedicated screen and readable Compose structure.	Implements
TripMonitoringScreen.kt	Uses a dedicated screen with clear naming and a simple UI structure.	Implements
ReportConcernScreen.kt	Uses a dedicated screen and clear navigation code.	Implements

3. Code That Does Not Fully Implement the Standards

File / Code	Finding	Assessment
LoginScreen.kt	The password visibility icon uses the same Android drawable for both visible and hidden states, so the UI does not clearly communicate the state.	Needs improvement
RegisterScreen.kt	The same visibility drawable is used for both states, similar to LoginScreen.kt.	Needs improvement
HomeScreen.kt	The Emergency button has an empty handler: // Handle emergency . A professional implementation should provide defined behavior or clearly mark the feature as unavailable.	Needs improvement
FareCalculationScreen.kt	The screen says This feature is coming soon! ; the main fare-calculation function is not implemented.	Incomplete
TripMonitoringScreen.kt	Route monitoring is presented as coming soon and does not yet perform actual monitoring.	Incomplete
TripHistoryScreen.kt	Trip history is presented as coming soon and does not yet load or display records.	Incomplete
ReportConcernScreen.kt	Report submission is not implemented; the screen only displays a placeholder.	Incomplete

4. Overall Finding

GeoFare generally follows professional coding practices through meaningful names, modular Compose screens, and organized navigation. However, several placeholder features and small UI implementation issues should be improved before the project can be considered fully compliant.